Fall, 2025

Deep Learning and Applications: HW-3
Due date: 2025/12/09 23:59

Instructions:

1. The use of **high-level APIs** (such as Keras, Slim, TFLearn, GluonTS, sktime, or darts) is **prohibited**. You are required to manually implement the forward computation of the LSTM. PyTorch or TensorFlow is recommended.

2. Submit your work as a single compressed file named: `HW3_StudentID.zip`. The submitted zip file must include the following:

   - **One Jupyter notebook** (`.ipynb`) containing your complete code and experimental results. The notebook should execute successfully from top to bottom without manual intervention.
   - **One report** (`.pdf`) explaining your implementation, experimental setup, results, and analysis in a clear and concise manner.

Air Quality Analysis and LSTM Forecasting

In this assignment, you will analyze real-world air-quality data from the **Taiwan Environmental Protection Administration (EPA)** and construct an LSTM-based model to forecast future PM2.5 values.

You will use the historical hourly measurements from the **Hsinchu monitoring station** during the **summer months of 2024 (June–August)**.

Dataset
Download the monthly CSV files for June, July, and August 2024 from the EPA open data portal:
`https://airtw.moenv.gov.tw/cht/Query/His_Data.aspx`

Your notebook must include the complete code for:

- Downloading and loading the three monthly CSV files.
- Filtering the records where `SiteName = 'Hsinchu'`.
- Cleaning missing values and normalizing numerical features.
- Constructing supervised learning sequences using sliding windows.

The following variables should be included in your analysis: PM2.5, PM10, O_3, NO_2, Temperature, Relative Humidity and Wind Speed.

Evaluation
Include the following items in your report and notebook:

- Correlation coefficient matrix of all selected features.

- Scatter plots (with marginal histograms) for at least one feature pair.
- Training and validation loss curves for your LSTM model.
- Prediction results on the test set, including at least five examples of:
  - 48-hour input sequence,
  - Ground-truth 24-hour PM2.5 future sequence,
  - Predicted 24-hour output sequence.
- Quantitative metrics: MAE, RMSE, and MAPE.

Task 1. Data Exploration
1. Download the summer 2024 EPA air-quality data and extract only the measurements from the Hsinchu station.
2. Compute the **correlation coefficient matrix** for all features and visualize it with a heatmap.
3. Select one feature pair (e.g., PM2.5 vs Temperature) and plot a scatter plot with marginal histograms.

Task 2. Sequence Construction
1. Using sliding windows, construct supervised training sequences:

- Input length: **48 hours**
- Output length: **next 24 hours of PM2.5**

2. Show the shapes of your dataset and visualize one complete input–output example.

Task 3. LSTM Forecasting Model
1. Build an LSTM model from scratch (no high-level forecasting libraries).
2. Train the model using a chronological split (70%/15%/15%).
3. Report evaluation results and visualize predictions as described above.
4. Briefly discuss whether the model performs better on short-term (1–6 hours) or long-term (18–24 hours) forecasts.

Figure 1: Steps for downloading the EPA 2024 Hsinchu air-quality data.

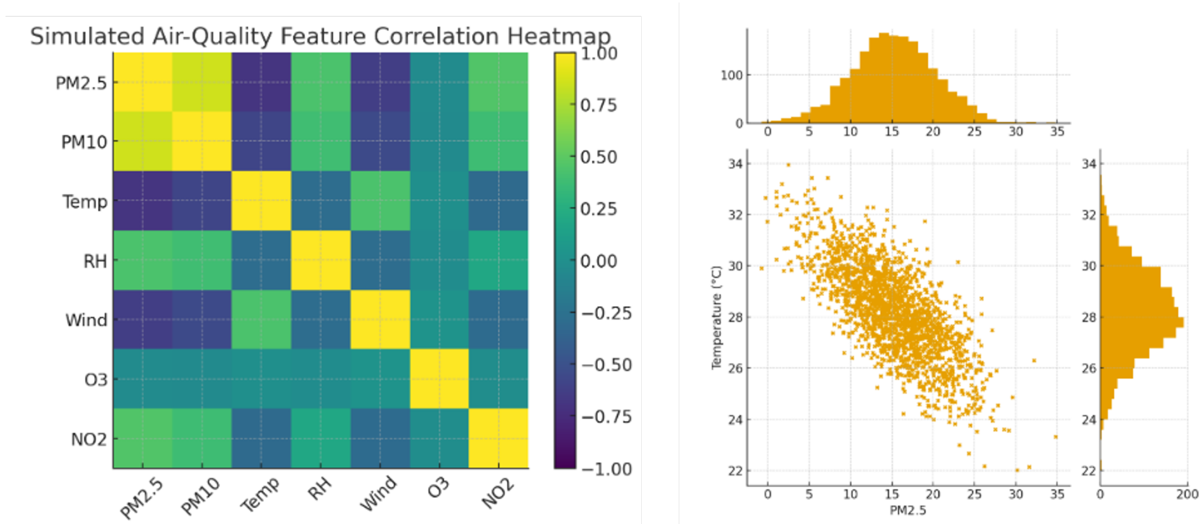

Figure 2: Example correlation heatmap and scatter plot visualization.

Lego Minifigure Generation

In this assignment, you will construct a **Variational Autoencoder (VAE)** to learn and synthesize images of Lego Minifigures.

You will use the `lego_minifigure_captions` dataset from Hugging Face (`https://huggingface.co/datasets/Armaggheddon/lego_minifigure_captions`). This dataset contains approximately 13,000 RGB images of full-body Lego Minifigures captured against clean white backgrounds, with most figures facing forward and a small portion shown at a 45° angle. The uniform background and consistent composition make this dataset suitable for generative modeling.

Task 1. Dataset Preparation and Exploration

1. Load the dataset using the Hugging Face `datasets` library and resize all images to a fixed resolution (e.g. 64×64). 2. Normalize all images and construct training/validation/test splits. 3. Visualize a grid of sampled Minifigure images to verify preprocessing correctness.

Task 2. VAE Training and Reconstruction

1. Implement a convolutional VAE consisting of:

- an encoder that outputs latent mean μ and log-variance $\log \sigma^2$,
- the reparameterization step

$$z = \mu + \sigma \odot \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

- a decoder that reconstructs images from latent vectors z.

2. Train the VAE using the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\mathrm{ELBO}} = \mathcal{L}_{\mathrm{reconstruction}} + \beta\, D_{\mathrm{KL}}\big(q(z \mid x) \,\|\, p(z)\big),$$

with $\beta = 1$ for the base model.
3. Include the following results in your report:

- Learning curves for the negative ELBO, reconstruction loss, and KL divergence.
- A grid of original images alongside their reconstructed outputs.
- A grid of novel samples generated by decoding latent vectors drawn from

$$z \sim \mathcal{N}(0, I).$$

Task 3. Latent Space Analysis and KL-Weight Experiments

1. Project the latent representations of test images into 2D using PCA. Discuss whether Minifigures with similar colors, accessories, or torso prints cluster together, and whether side-facing figures form a separate region.

2. Select two Minifigure images x_a and x_b, encode them to z_a and z_b, and generate a latent interpolation:

$$z(\alpha) = (1 - \alpha)\, z_a + \alpha\, z_b, \qquad \alpha \in [0, 1].$$

Visualize and analyze how attributes change along the interpolation path.

3. Train three VAE variants with KL weights $\beta = 0,\ 1,\ 100$. Compare the following aspects:

- reconstruction sharpness,
- generative diversity,
- latent space smoothness.

Provide a concise discussion of the trade-offs observed among these three models.

Figure 3: Samples from Lego Minifigure images.